Translating situations into algebraic expressions

1 If Sam is currently x years old, write an expression using $x$to represent his age after 5 years.

2 Write an expression for each of the following:

a 8 more than $-3x$

b 10 less than $8x$

c The product of $9u$ and 3

d One-seventh of $4x$

3 If the distance of a marathon is $10x$ m, what would be the distance of a half marathon?

4 If y represents the number of houses then write an expression for 10 less than 16 times the number of houses.

5 If x represents the number of helmets then write an expression for 6 less than 18 divided by the number of helmets .

6 Fiona ran 9 miles more than Katrina in a training session.
Let v be number of miles run by Katrina. Write an algebraic expression for the number of miles run by Fiona.

7 The height of a flag pole is one-third of the height of a building.
Let t be height of building. Write an algebraic expression for the height of flag pole.

8 The width of a rectangle is 19 cm less than double the length.
Let n be length of the rectangle. Write an algebraic expression for the width.

9 Write an algebraic expression for each of the following phrases. Let the variable x represent the number.

a "Four times a number, decreased by 3"

b "Eight more than the quotient of 9 and a number"

c "The sum of 9 divided by a number and that number divided by 9"

d "The sum of 2 and x is multiplied by 5. The result is then taken away from 15."

10 Jill is 9 cm taller than Ally, while Melanie is 2 cm shorter than Ally.
Let v be Ally's height in centimeters.

a Write an algebraic expression for Jill's height.

11 Emma has $\$80$ in her wallet. If she spends $\$m$, how much will she have remaining?

12 Side B of a triangle is 9 cm longer than side A. Side C is 8 cm shorter than three times the length of side A.
Let y be length of side A.

a Write an alegbraic expression for the length of side B.

b Write an algebraic expression for the length of side C.

13 The width of a rectangular football field is 8 m more than half of its length. If L represents the length of the field, write an expression for the width.

14 Tara is with a mobile phone provider that charges $\$0.24$ per minute plus a connection fee of $\$0.35$. How much will she be charged for a call that lasts q minutes?

15 Judy saved some of her money in a bank for one year. In this time, she earned $\$90$ interest. At the end of the year, she withdrew $\$85$ to spend. If the amount Judy initially saved is given by x, write an expression for the amount she has left.

16 Mohamad has three balls; a ping pong ball, a golf ball and a tennis ball. The golf ball weighs 18 times as much as the ping pong ball and the tennis ball weighs 12 grams more than the golf ball.

a If the ping pong ball weighs x grams, write an algebraic expression for the weight of the golf ball.

b Now write an algebraic expression for the weight of the tennis ball.

17 If x represents the cost of one orange and y represents the cost of one onion, build an expression for the cost of 5 oranges and 9 onions.

18 A gas heating system costs $\$23\,700$ to install and has operating costs of $\$255$ per year. Write an expression for the total cost of the gas heating system after x years.

19 A rower is rowing at a speed of $(x + y) \text{ km/h}$ with the current. Write an expression for the total distance she will cover after 4 hours.

20 The table shows the cost of certain items in a vending machine. If Laura buys v packets of chewing gum and 4 packets of jelly beans, how much does she have to pay in total?

Item	Cost
jelly beans	$2.70
chewing gum	$3.80
cookies	$1.10
chips	$3.60
Roasted peanuts	$1.40

21 The first of two consecutive numbers is $\dfrac{7+p}{2}$. Write an expression for the other number.

22 The first of two consecutive even numbers is $\dfrac{x}{4} - 9$. Write an expression for the next even number.

23 If a triangle has one side of length $4x$ cm, another double this length, and the final side being 8 cm, write an expression for its perimeter.

24 Translate the following expression in words:

$$\frac{5}{p+2} - 3$$

25 Tobias has three times as many books as Hermione does, and Hermione has four less than triple the number of books that Dave does.
If Dave has n books, write an expression for the number of books that Tobias has.

26 In a multiple choice quiz, 3 points are awarded for a right answer, and 2 points are deducted for a wrong answer. If Luigi got x questions right and y questions wrong, write an expression for how many points he got.

27 A triangle has side lengths of $9x + 3$ cm, $10x - 3y$ cm , and $12 - 4y$ cm.

a Write an expression for the perimeter of the triangle.

b Find the perimeter of the triangle when $x = 9$ and $y = 5$.

28 Each week, Aaron earns $\$m$ more than Paul. Uther earns $\$u$ more than Aaron. If Paul earns $\$440$ per week, how much does Uther earn per week?

29 The local stationery shop has a sale on boxes of pens. Each box contains an unknown number of pens.
Hermione bought 13 boxes of pens. Caitlin bought 12 boxes of pens.
Write an expression for the total number of pens the two girls bought.
Use the variable p to represent the number of pens in each box.

30 Glen's weekly pocket money of $\$z$ increased by 9%.

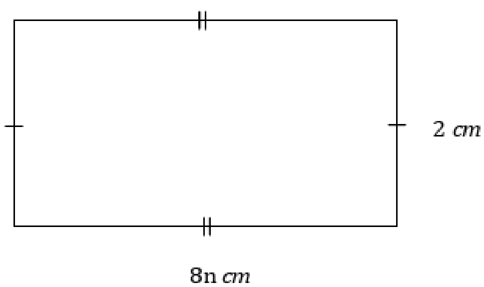
a How much will he receive per week after the increase? Your answer should involve a decimal.

b If Glen's original pocket money amount was $\$20$, what is his new pocket money amount?

31 Two friends Jenny and Kathleen share $\$65$ in the ratio $w : y$ where w and y are both positive and $w < y$.
Write an expression for the amount of money Jenny receives.

32 Write an expression for the perimeter of the the following shapes:

a



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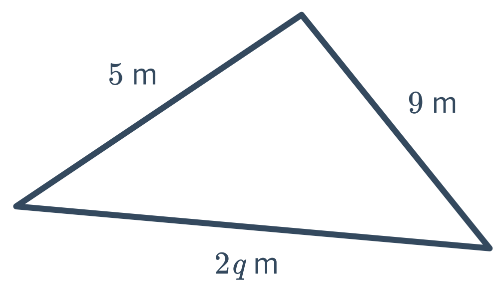

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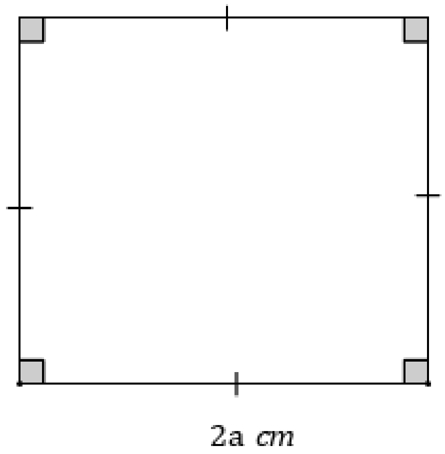
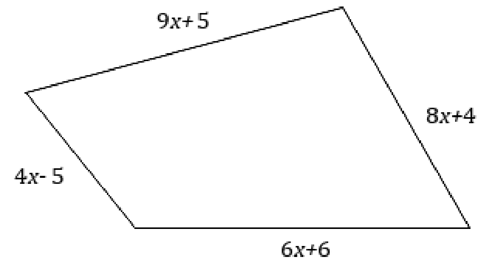
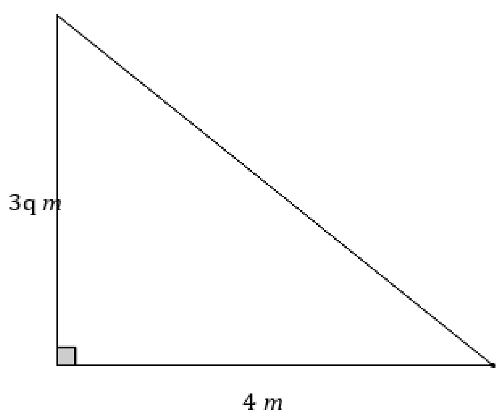




33 Write an expression for the area of the following figures:

a



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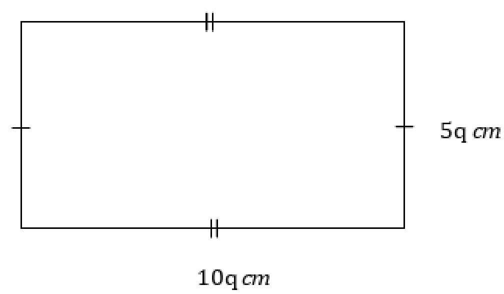

34 Glen lived in a rectangular bedroom. His mother asked him to find out the area of his room, and since Glen did not have a measuring tape or ruler, he counted how many of his steps each wall was.
If Glen's foot length is x cm, and two adjacent walls were 12 and 10 footsteps long, write an expression for the area of his room.

35 John’s yearbook committee is searching for a printing company. A particular company charges $\$130$ for a onetime set-up fee and $\$7$ for each printed yearbook.
Write an expression representing the total cost, in dollars, of x number of yearbooks for John's school.

Additional questions

36 Dylan purchased 3 pens, 4 pencils and a single note pad. The total cost of all these items was $3x+4y+5$ dollars.

a What does the variable y represent?

b What does the variable x represent?

c What does 5 represent?

37 Deborah earns $\$25$ per hour of work and is paid double for every hour worked on the weekend. At the end of a week, Deborah calculates her pay for that week to be $25x+50y$ dollars.

a What represents the number of hours Deborah worked during the weekdays?

b What represents the number of hours Deborah worked during the weekends?

38 Mohamad and Valentina run a clothes business. Mohamad is able to sew 5 shirts an hour, while Valentina is able to sew 3 shirts an hour. The expression $6\left(5+3\right)$ represents the total number of shirts the couple is able to produce in a six-hour period.
What does $5+3$ represent?

39 Brad and Patricia are making paper cranes to decorate their room. Brad can make m an hour while Patricia can make n an hour. Brad spends 6 hours making cranes while Patricia spends 5 hours over the weekend.
Together they make a total of $6m+5n$ paper cranes over the weekend.

a What does $6m$represent?

b What does $5n$ represent?

40 The width of a rectangular football field is 7 yards more than half of its length. If L represents the length of the field, write an expression for the width.